1. PackageQuantityDiscountCalculator

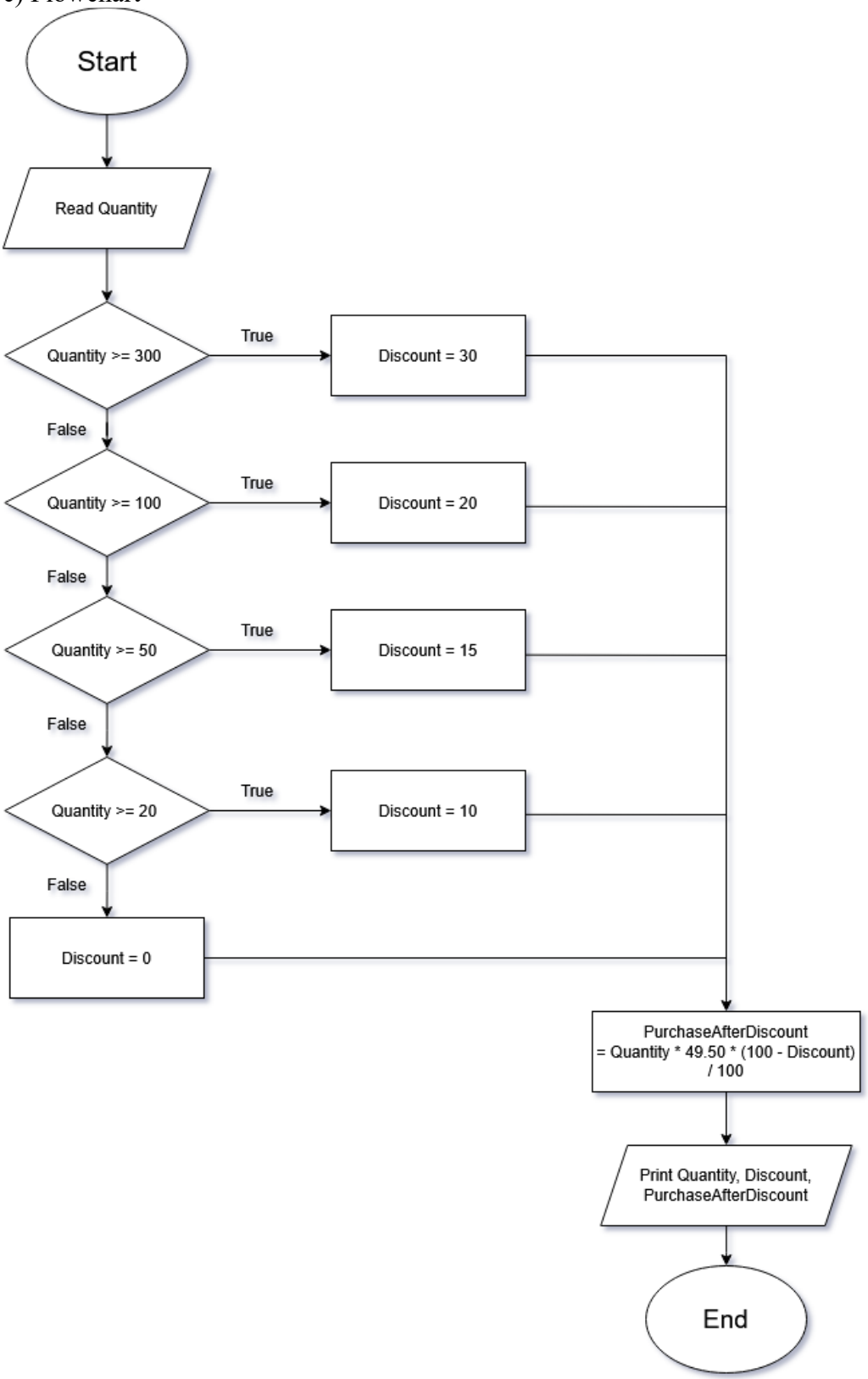
a) Problem Analysis Chart (PAC)

Given Data	Required Results
1. Quantity of packages purchased	1. Package quantity 2. Amount of the discount 3. Total amount of the purchase after the discount
Processing Required	**Solution Alternative**
Discount = if(Quantity in some range) then (return CertainDiscount) PurchaseAfterDiscount = Quantity * 49.50 * (100 - Discount) / 100	1. Define the price of single package as constants. 2. Define the quantity of packages purchased as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
Quantity of packages purchased	1. Enter Quantity 2. If (Quantity >= 300,100,50,20) Discount = 30,20,15,10 else Discount = 0 3. Calculate PurchaseAfterDiscount 4. Print Quantity 5. Print Discount 6. Print PurchaseAfterDiscount 7. End	-	1. Package quantity 2. Amount of the discount 3. Total amount of the purchase after the discount

c) Flowchart


Start
Read Quantity
Quantity >= 300
True
Discount = 30
False
Quantity >= 100
True
Discount = 20
False
Quantity >= 50
True
Discount = 15
False
Quantity >= 20
True
Discount = 10
False
Discount = 0
PurchaseAfterDiscount
= Quantity * 49.50 * (100 - Discount)
/ 100
Print Quantity, Discount,
PurchaseAfterDiscount
End

d) Pseudocode

```
Start
        Read Quantity
        if (Quantity >= 300)
                Discount = 30
        else if (Quantity >= 100)
                Discount = 20
        else if (Quantity >= 50)
                Discount = 15
        else if (Quantity >= 20)
                Discount = 10
        else
                Discount = 0
        PurchaseAfterDiscount = Quantity * (100 - Discount) / 100
        Print Quantity, Discount, PurchaseAfterDiscount
End
```

e) Java code

```java
import java.util.Scanner;

public class PackageQuantityDiscountCalculator {
        public static void main(String[] args){
                // Get quantity of package purchased from user
                System.out.println("Please enter the number of number of packages purchased (Each package is RM49.50): ");
                Scanner scanner = new Scanner(System.in);
                int quantity = scanner.nextInt();

                // Check discount
                int discount;
                if (quantity >= 300){
                        discount = 30;
                } else if (quantity >= 100) {
                        discount = 20;
                } else if (quantity >= 50) {
                        discount = 15;
                } else if (quantity >= 20) {
                        discount = 10;
                } else {
                        discount = 0;
                }

                // Calculate purchase after discount
                final float priceOfSinglePackage = 49.50f;
                float purchaseAfterDiscount = quantity * priceOfSinglePackage * (100 - discount) / 100;

                // Print output
                System.out.println("Number of packages purchased : " + quantity);
                System.out.println("Discount : " + discount + "%");
                System.out.println("Price after discount : RM" + String.format("%.2f",purchaseAfterDiscount));
        }
}
```

2. MenuInRestaurant

a) Problem Analysis Chart (PAC)

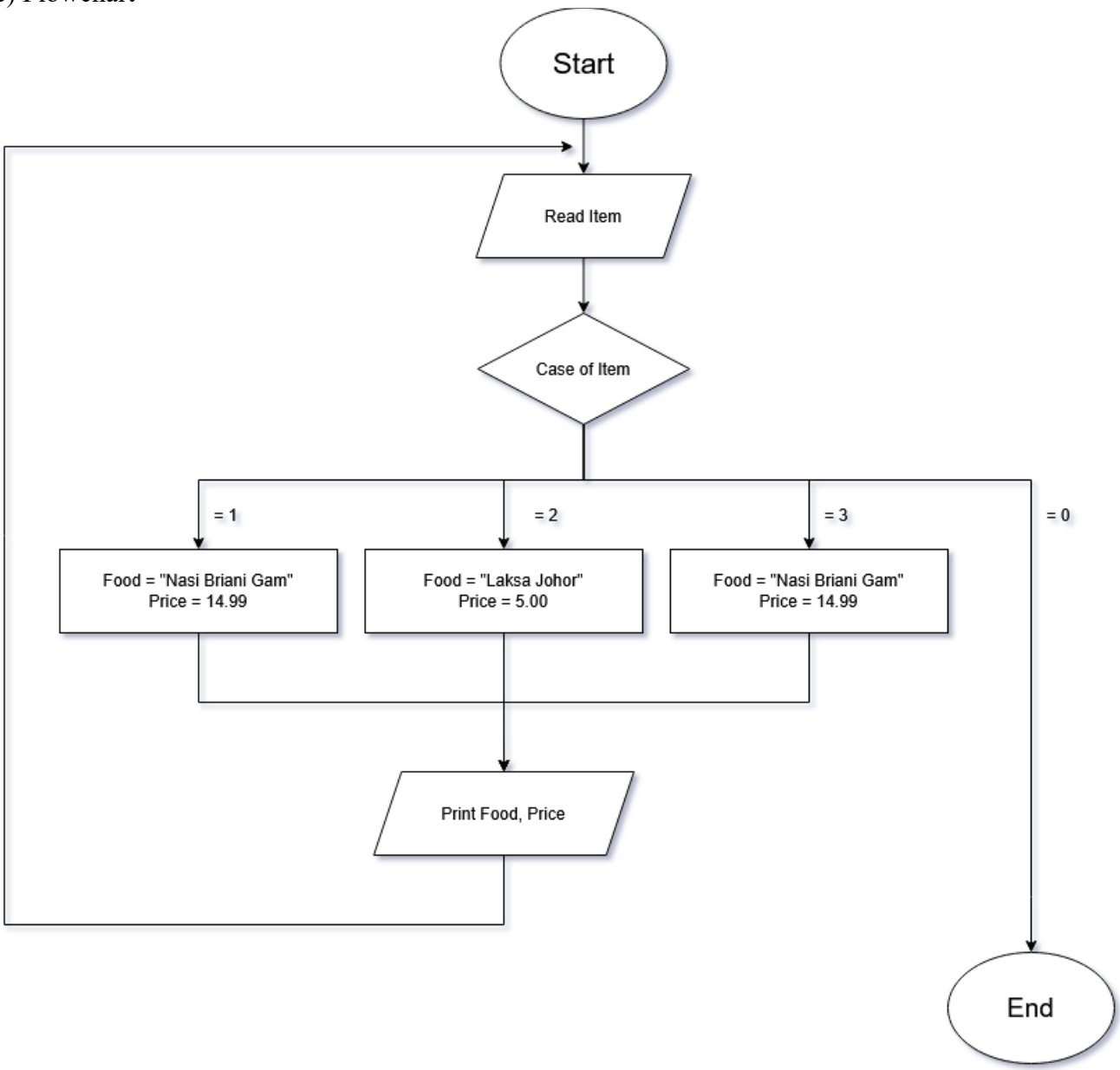
Given Data	Required Results
1. Item	1. Selected food 2. Price
Processing Required	**Solution Alternative**
switch (item) case 1,2,3: display food, price case 0: quit repeat the process until user quit	1. Define item as constants. 2. Define item as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output

Item	1. Enter Item
	2. switch (Item)
	 case 0:
	  jump to step 3
	 case 1,2,3:
	  print food, price
	  jump to step 1
	3. End
		-	1. Selected food
			2. Price

c) Flowchart



d) Pseudocode

```
Start
        Read Item
        switch(Item)
                case 0:
                        Jump to End
                case 1:
                        Food = "Nasi Briani Gam"
                        Price = 14.99
                        break
                case 2:
                        Food = "Laksa Johor"
```

Price = 5.00
break
case 3:
Food = "Nasi +2 lauk"
Price = 6.00
break
Print Food, Price
Jump to Start
End

e) Java code

```java
import java.util.Scanner;
import java.util.InputMismatchException;

public class MenuInRestaurant {

        public static Scanner scanner = new Scanner(System.in);
        public static int item;
        public static String food;
        public static float price;
        public static int input;

        public static void main(String[] args){
                // Print welcome message
                System.out.println("Welcome to the restaurant!");
                System.out.println("");

                // Loop
                while (true){
                        // Get item number from user
                        item = getItem();
                        // Set food and price based on item selected
                        setFoodAndPrice(item);

                        // Print selected food and price
                        System.out.println("");
                        System.out.println("Item : " + item);
                        System.out.println("Selected food : " + food);
                        System.out.println("Price : RM" + String.format("%.2f",price));
                        System.out.println("");

                        // Tell user to select for other item
                        System.out.println("You can select from other items.");
                        System.out.println("");
                }
        }

        public static int getItem(){
                System.out.println("Please choose a number from the item lists : ");
                System.out.println("Type 1,2,3 for foods");
                System.out.println("Type 0 to quit");
                input = -1;
                try {
                        input = scanner.nextInt();
                } catch (InputMismatchException e){
                        System.out.println("Please enter a number.");
                        System.out.println("");
                        scanner.nextLine();
                        getItem();
                }
                if (input > 3 || input < 0){
                        System.out.println("You enter a wrong number.");
                        System.out.println("");
                        getItem();
                }
                return input;
        }

        public static void setFoodAndPrice(int item){
                switch (item){
                        case 0:
                                System.out.println("");
                                System.out.println("Thanks for coming!");
                                System.exit(0);
                        case 1:
                                food = "Nasi Briani Gam";
                                price = 14.99f;
                                break;
                        case 2:
                                food = "Laksa Johor";
                                price = 5.00f;
                                break;
                        case 3:
                                food = "Nasi +2 lauk";
                                price = 6.00f;
                                break;
                }
        }
}
```

3. GradeCalculator

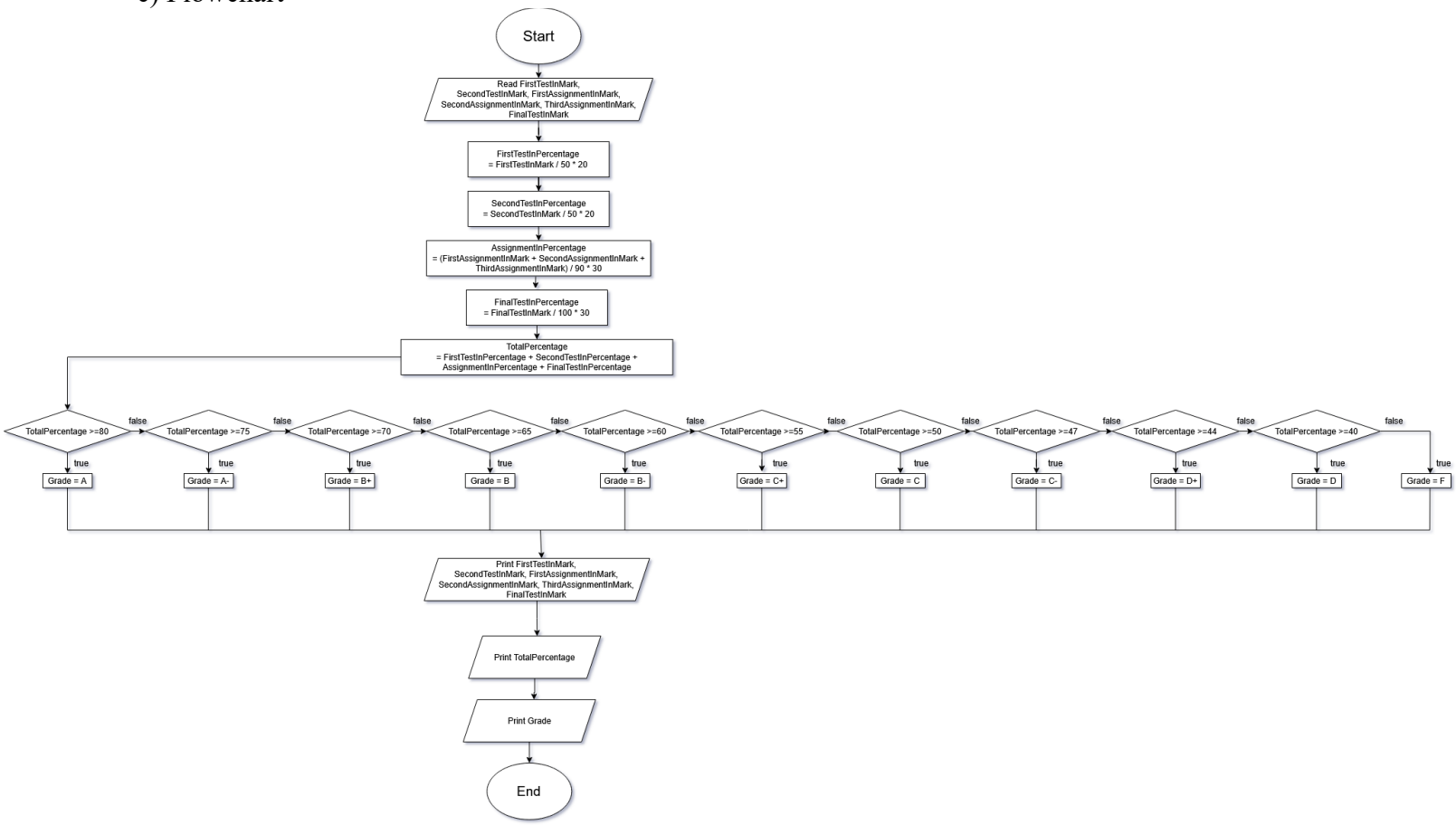
a) Problem Analysis Chart (PAC)

Given Data	Required Results
1. Mark for first test 2. Mark for second test 3. Mark for assignments 4. Mark for final exam	1. All marks in raw mark 2. Total mark in percentage 3. Grade
Processing Required	**Solution Alternative**
FirstTestInPercentage = FirstTestInMark / 50 * 20 SecondTestInPercentage = SecondTestInMark / 50 * 20 AssignmentInPercentage = (FirstAssignmentInMark + SecondAssignmentInMark + ThirdAssignmentInMark) / 90 * 30 FinalTestInPercentage = FinalTestInMark / 100 * 30 TotalPercentage = FirstTestInPercentage + SecondTestInPercentage + AssignmentInPercentage + FinalTestInPercentage if(TotalPercentage in some ranges) Grade = A/A-/B+/B/B-/C+/C/C-/D+/D/F	1. Define marks for each tests as constants. 2. Define marks for each tests as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
1. FirstTest 2. SecondTest 3. First Assignment 4. Second Assignment 5. Third Assignment 6. FinalTest	1. Enter FirstTest, SecondTest, FirstAssignment, SecondAssignment, ThirdAssignment, FinalTest in mark form 2. Calculate FirstTest, SecondTest, Assignment, FinalTest in percentage form 3. Calculate TotalPercentage 4. If TotalPercentage in ranges, Calculate Grade 5. Display FirstTest, SecondTest, FirstAssignment, SecondAssignment, ThirdAssignment, FinalTest 6. Display TotalPercentage 7. Display Grade 8. End	-	1. Raw marks for each tests 2. TotalPercentage 3. Grade

c) Flowchart



d) Pseudocode

```
Start
        Read {
        FirstTestInMark, SecondTestInMark, FirstAssignmentInMark,
        SecondAssignmentInMark, ThirdAssignmentInMark, FinalTestInMark
        }
        FirstTestInPercentage = FirstTestInMark / 50 * 20
        SecondTestInPercentage = SecondTestInMark / 50 * 20
        AssignmentInPercentage = {
        (FirstAssignmentInMark + SecondAssignmentInMark + ThirdAssignmentInMark) / 90 * 30
        }
        FinalTestInPercentage = FinalTestInMark / 100 * 30
        TotalPercentage = {
        FirstTestInPercentage + SecondTestInPercentage +
        AssignmentInPercentage + FinalTestInPercentage
        }
        if (TotalPercentage >= 80)
                Grade = A
        else if (TotalPercentage >= 75)
                Grade = A-
```

```
        else if (TotalPercentage >= 70)
                Grade = B+
        else if (TotalPercentage >= 65)
                Grade = B
        else if (TotalPercentage >= 60)
                Grade = B-
        else if (TotalPercentage >= 55)
                Grade = C+
        else if (TotalPercentage >= 50)
                Grade = C
        else if (TotalPercentage >= 47)
                Grade = C-
        else if (TotalPercentage >= 44)
                Grade = D+
        else if (TotalPercentage >= 40)
                Grade = D
        else
                Grade = F
        Print
        {
        FirstTestInMark, SecondTestInMark, FirstAssignmentInMark, SecondAssignmentInMark,
        ThirdAssignmentInMark, FinalTestInMark
        }
        Print TotalPercentage
        Print Grade
End
```

e) Java code

```java
import java.util.Scanner;
import java.util.InputMismatchException;

public class GradeCalculator{

        public static Scanner scanner = new Scanner(System.in);
        public static int input;

        public static void main(String[] args){
                // Print welcome message
                System.out.println("Welcome to grade calculator!");

                // Init variables
                String[] markMessages = {
                        "First test (Full mark is 50) : ",
                        "Second test (Full mark is 50) : ",
                        "First assignment (Full mark is 30) : ",
                        "Second assignment (Full mark is 30) : ",
                        "Third assignment (Full mark is 30) : ",
                        "Final test (Full mark is 100) : "
                };
                int[] fullMarks = {50,50,30,30,30,100};
                int[] markToPercentage = {20,20,30,30};
                int[] marks = new int[6];
                int[] percentages = new int[4];

                // Get raw marks from user
                System.out.println("Please enter the marks for each tests below.");
                boolean success;
                for (int i = 0;i<6;i++){
                        do{
                                System.out.println(markMessages[i]);
                                success = getAndCheckMark(fullMarks[i]);
                                if (success){
                                        marks[i] = input;
                                }
                        } while (!success);
                }
```

```java
        // Convert marks to percentage form
        // First test
        percentages[0] = marks[0] * markToPercentage[0] / fullMarks[0];
        // Second test
        percentages[1] = marks[1] * markToPercentage[1] / fullMarks[1];
        // Assigments
        percentages[2] = (marks[2] + marks[3] + marks[4]) * markToPercentage[2] / (fullMarks[2] + fullMarks[3] + fullMarks[4]);
        // Final test
        percentages[3] = marks[5] * markToPercentage[3] / fullMarks[5];

        // Calculate total percentage
        int totalPercentage = percentages[0] + percentages[1] + percentages[2] + percentages[3];
         // Find grade
         String grade;
         if (totalPercentage >= 80)
                 grade = "A";
         else if (totalPercentage >= 75)
                 grade = "A-";
         else if (totalPercentage >= 70)
                 grade = "B+";
         else if (totalPercentage >= 65)
                 grade = "B";
         else if (totalPercentage >= 60)
                 grade = "B-";
         else if (totalPercentage >= 55)
                 grade = "C+";
         else if (totalPercentage >= 50)
                 grade = "C";
         else if (totalPercentage >= 47)
                 grade = "C-";
         else if (totalPercentage >= 44)
                 grade = "D+";
         else if (totalPercentage >= 40)
                 grade = "D";
         else
                 grade = "F";
```

```java
            // Print outputs
            System.out.println("");
            System.out.println("Result : ");
            System.out.println("");
            for (int i = 0;i<6;i++){
                    System.out.println(markMessages[i] + "" + marks[i]);
            }
            System.out.println("");
            System.out.println("Total percentage : " + totalPercentage + "%");
            System.out.println("Grade : " + grade);
    }

    public static boolean getAndCheckMark(int fullMark){
            try {
                    input = scanner.nextInt();
            } catch (InputMismatchException e){
                    System.out.println("Please enter an integer.");
                    scanner.nextLine();
                    return false;
            }
            if (input > fullMark || input < 0){
                    System.out.println("You enter a wrong number. Please try again.");
                    return false;
            }
            return true;
    }
}
```

4. VendorDiscountCalculator

a) Problem Analysis Chart (PAC)

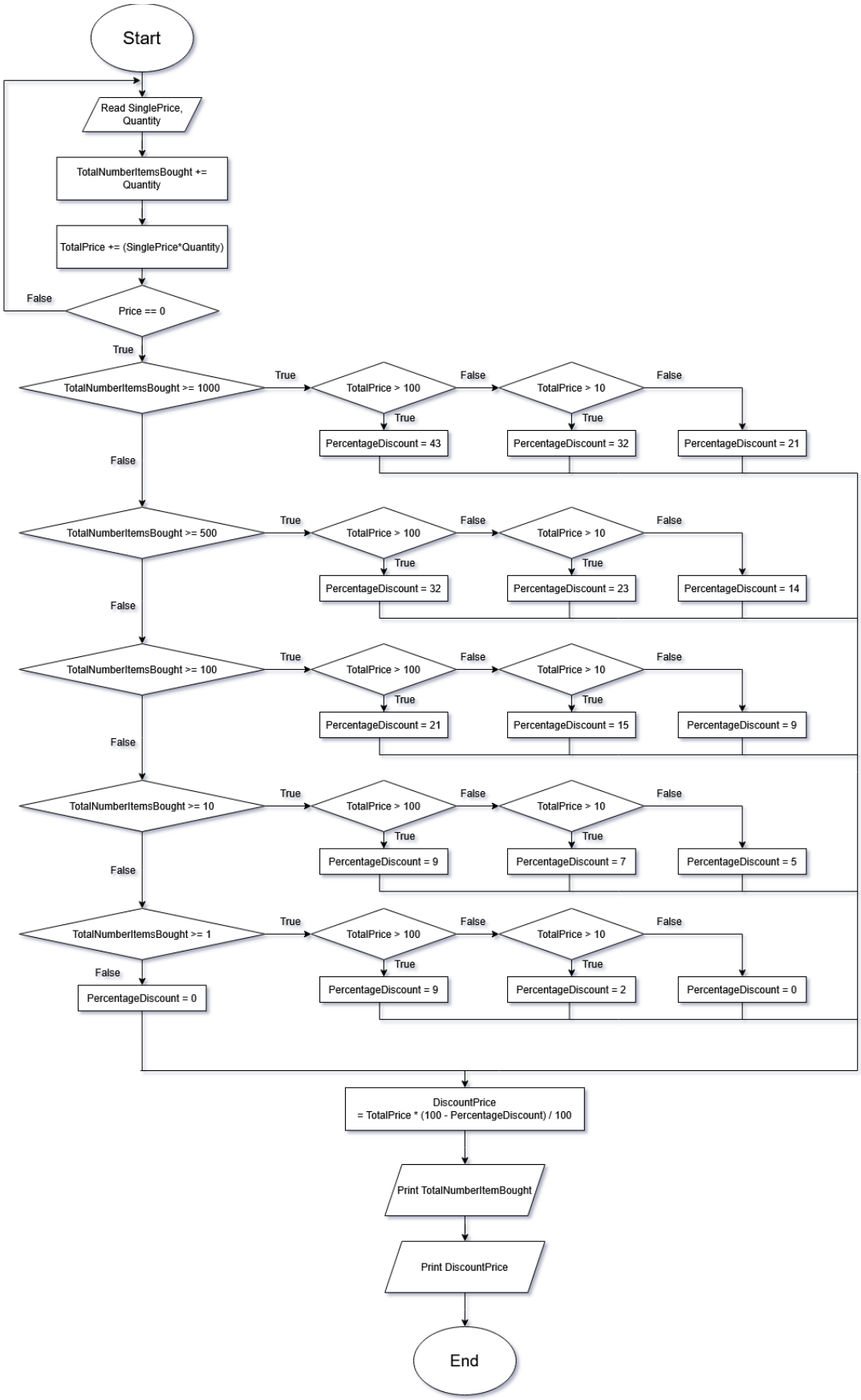
Given Data	Required Results
1. Each items prices 2. Quantity of each item	1. Total unit 2. Total price after discount
Processing Required	**Solution Alternative**

TotalNumberItemsBought += Quantity, after each items added
 	
TotalPrice += (SinglePrice*Quantity), after each items added
 	
if (TotalNumberItemsBought in some ranges){
 if (TotalPrice in some ranges){
  set PercentageDiscount
 }
}
 	
DiscountPrice = {
TotalPrice * (100 - PercentageDiscount) / 100
}
	1. Define prices for each items, quantity of each item as constants.
	2. Define prices for each items, quantity of each item as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
1. Each items prices 2. Quantity of each item	1. Enter prices for each items 2. Enter quantity for each items 3. Track TotalNumberItemsBought 4. Track TotalPrice 5. If TotalNumberItemsBought in ranges, If TotalPrice in some ranges, Calculate PercentageDiscount 6. Calculate DiscountPrice 7. Display TotalNumberItemsBought 8. Display DiscountPrice 9. End	-	1. TotalNumberItems Bought 2. DiscountPrice

c) Flowchart


Start
Read SinglePrice, Quantity
TotalNumberItemsBought += Quantity
TotalPrice += (SinglePrice*Quantity)
Price == 0
False
True
TotalNumberItemsBought >= 1000
TotalPrice > 100
PercentageDiscount = 43
TotalPrice > 10
PercentageDiscount = 32
PercentageDiscount = 21
TotalNumberItemsBought >= 500
PercentageDiscount = 32
PercentageDiscount = 23
PercentageDiscount = 14
TotalNumberItemsBought >= 100
PercentageDiscount = 21
PercentageDiscount = 15
PercentageDiscount = 9
TotalNumberItemsBought >= 10
PercentageDiscount = 9
PercentageDiscount = 7
PercentageDiscount = 5
TotalNumberItemsBought >= 1
PercentageDiscount = 9
PercentageDiscount = 2
PercentageDiscount = 0
PercentageDiscount = 0
DiscountPrice = TotalPrice * (100 - PercentageDiscount) / 100
Print TotalNumberItemBought
Print DiscountPrice
End

d) Pseudocode

```
Start
        while (SinglePrice != 0){
                Read SinglePrice, Quantity
                TotalNumberItemsBought += Quantity
                TotalPrice += SinglePrice*Quantity
        }
        if (TotalNumberItemsBought >= 1000)
                if (TotalPrice > 100)
                        PercentageDiscount = 43
                else if (TotalPrice > 10)
                        PercentageDiscount = 32
                else
                        PercentageDiscount = 21
        else if (TotalNumberItemsBought >= 500)
                if (TotalPrice > 100)
                        PercentageDiscount = 32
                else if (TotalPrice > 10)
                        PercentageDiscount = 23
                else
                        PercentageDiscount = 14
        else if (TotalNumberItemsBought >= 100)
                if (TotalPrice > 100)
                        PercentageDiscount = 21
                else if (TotalPrice > 10)
                        PercentageDiscount = 15
                else
                        PercentageDiscount = 9
        else if (TotalNumberItemsBought >= 10)
                if (TotalPrice > 100)
                        PercentageDiscount = 9
                else if (TotalPrice > 10)
                        PercentageDiscount = 7
                else
                        PercentageDiscount = 5
        else if (TotalNumberItemsBought >= 1)
                if (TotalPrice > 100)
                        PercentageDiscount = 5
                else if (TotalPrice > 10)
                        PercentageDiscount = 2
                else
                        PercentageDiscount = 0
        else
                PercentageDiscount = 0
        DiscountPrice = TotalPrice * (100 - PercentageDiscount) / 100
        Print TotalNumberItemsBought
        Print DiscountPrice
End
```

e) Java code

```java
public class VendorDiscountCalculator{

    public static Scanner scanner = new Scanner(System.in);
    public static float inputPrice;
    public static int inputQuantity;
    public static boolean success;
    public static float totalPrice;
    public static int totalNumberItemsBought;

    public static void main(String[] args){
        // Print welcome message
        System.out.println("Welcome to vendor discount calculator!");
        System.out.println("Please input all the items bought one by one.");
        System.out.println("");

        // Get list of items from user
        float singlePrice;
        int quantity;
        int counter = 1;
        while (true){
            System.out.println("Item no. " + counter);

            System.out.println("Enter the price (RM) : ");
            do{
                singlePrice = getPrice();
            } while(!success);
            if (singlePrice == 0) break;

            System.out.println("Enter the quantity : ");
            do{
                quantity = getQuantity();
            } while(!success);

            // Track total items bought and total price
            totalNumberItemsBought += quantity;
            totalPrice += (singlePrice*quantity);

            counter++;

            System.out.println("");
        }
```

```java
        // Calculate percentage discount
        int percentageDiscount;
        if (totalNumberItemsBought >= 1000){
            if (totalPrice > 100)
                percentageDiscount = 43;
            else if (totalPrice > 10)
                percentageDiscount = 32;
            else
                percentageDiscount = 21;
        } else if (totalNumberItemsBought >= 500){
            if (totalPrice > 100)
                percentageDiscount = 32;
            else if (totalPrice > 10)
                percentageDiscount = 23;
            else
                percentageDiscount = 14;
        } else if (totalNumberItemsBought >= 100){
            if (totalPrice > 100)
                percentageDiscount = 21;
            else if (totalPrice > 10)
                percentageDiscount = 15;
            else
                percentageDiscount = 9;
        } else if (totalNumberItemsBought >= 10){
            if (totalPrice > 100)
                percentageDiscount = 9;
            else if (totalPrice > 10)
                percentageDiscount = 7;
            else
                percentageDiscount = 5;
        } else if (totalNumberItemsBought >= 1){
            if (totalPrice > 100)
                percentageDiscount = 5;
            else if (totalPrice > 10)
                percentageDiscount = 2;
            else
                percentageDiscount = 0;
        } else
            percentageDiscount = 0;
```

```java
        // Calculate discounted price
        float discountPrice = totalPrice * (100 - percentageDiscount) / 100;

        // Print outputs
        System.out.println("");
        System.out.println("Total items bought : " + totalNumberItemsBought);
        System.out.println("Discounted Price : RM" + String.format("%.2f",discountPrice));
    }

    public static float getPrice(){
        try {
            inputPrice = scanner.nextFloat();
        } catch (InputMismatchException e){
            System.out.println("Please enter a correct price.");
            scanner.nextLine();
            success = false;
            return -1;
        }
        if (inputPrice < 0) {
            System.out.println("Please enter a positive number.");
            success = false;
            return -1;
        }
        success = true;
        return inputPrice;
    }

    public static int getQuantity(){
        try {
            inputQuantity = scanner.nextInt();
        } catch (InputMismatchException e){
            System.out.println("Please enter an integer.");
            scanner.nextLine();
            success = false;
            return -1;
        }
        if (inputQuantity < 0) {
            System.out.println("Please enter a positive number.");
            success = false;
            return -1;
        }
        success = true;
        return inputQuantity;
    }
}
```

5. DynamicTieredBillingSystem

a) Problem Analysis Chart (PAC)

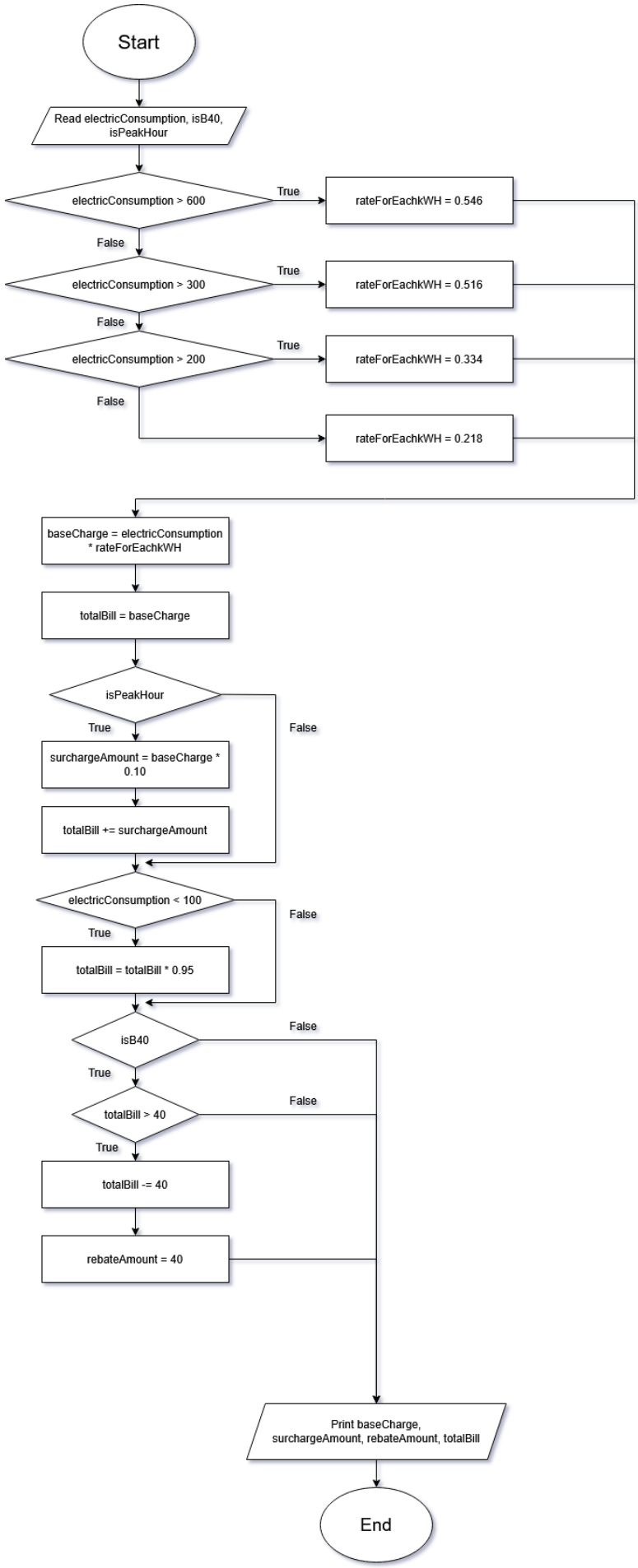
Given Data	Required Results
1. Electric consumption 2. User category 3. Peak Hour surcharge	1. Base Charge 2. Surcharge Amount 3. Rebate Amount 4. Final Total Bill
Processing Required	**Solution Alternative**

if (electricConsumption in some ranges)
  set rateForEachkWH
baseCharge= electricConsumption *
rateForEachkWH
totalBill = baseCharge
 	
if (isPeakHours)
  surchargeAmount = totalBill * 0.10
  totalBill += surchargeAmount
 	
if (electricConsumption < 100)
  totalBill = totalBill * 0.95
 	
rebateAmount = 40
if (isB40)
  if (totalBill > rebateAmount)
    totalBill -= rebateAmount
	1. Define prices for electric consumption, user category and peak hour surcharge as constants.
	2. Define electric consumption, user category and peak hour surcharge as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
1. Electric consumption 2. User category 3. Peak Hour surcharge	1. Enter electricConsumption 2. Enter isB40 3. Enter isPeakHour 4. If electricConsumption in ranges, Set rateForEachkWH 5. Calculate baseCharge 6. Calculate surchageAmount 7. Calculate discount 8. Calculate rebateAmount 9. Calculate totalBill 10. Display baseCharge, surchargeAmount, rebateAmount, totalBill 11. End	-	1. Base Charge 2. Surcharge Amount 3. Rebate Amount 4. Final Total Bill

c) Flowchart


Start
Read electricConsumption, isB40, isPeakHour
electricConsumption > 600
True
rateForEachkWH = 0.546
False
electricConsumption > 300
True
rateForEachkWH = 0.516
False
electricConsumption > 200
True
rateForEachkWH = 0.334
False
rateForEachkWH = 0.218
baseCharge = electricConsumption * rateForEachkWH
totalBill = baseCharge
isPeakHour
True
False
surchargeAmount = baseCharge * 0.10
totalBill += surchargeAmount
electricConsumption < 100
True
False
totalBill = totalBill * 0.95
isB40
True
False
totalBill > 40
True
False
totalBill -= 40
rebateAmount = 40
Print baseCharge, surchargeAmount, rebateAmount, totalBill
End

d) Pseudocode

```
Start
        Read electricConsumption, isB40. isPeakHour
        if (electricConsumption > 600)
                rateForEachkWH = 0.546
        else if (electricConsumption > 300)
                rateForEachkWH = 0.516
        else if (electricConsumption > 200)
                rateForEachkWH = 0.334
        else
                rateForEachkWH = 0.218
        baseCharge = electricConsumption * rateForEachkWH
        totalBill = baseCharge
        if (isPeakHour)
                surchargeAmount = baseCharge * 0.10
                totalBill += surchargeAmount
        if (electricConsumption < 100)
                totalBill = totalBill * 0.95
        if (isB40)
                if (totalBill > 40)
                        totalBill -= 40
                        rebateAmount = 40
        Print baseCharge,surchargeAmount, rebateAmount, totalBill
End
```

e) Java code

```java
import java.util.Scanner;
import java.util.InputMismatchException;

public class DynamicTieredBillingSystem{

        public static Scanner scanner = new Scanner(System.in);
        public static boolean success;

        public static void main(String[] args){
                // Print welcome message
                System.out.println("Welcome to Dynamic Electricity Billing & Sustainability Incentive system!");
                System.out.println("");

                // Init variables
                int electricConsumption;
                boolean isB40,isPeakHour;
                double totalBill = 0;
                double rateForEachkWh;

                // Get inputs
                System.out.println("Enter electric consumption (kWh) : ");
                do{
                        electricConsumption = getElectricConsumption();
                } while (!success);

                System.out.println("Enter user category (1 for B40, 2 for Standard): ");
                do{
                        isB40 = getB40();
                } while (!success);

                System.out.println("Is it peak hour (1 for Yes, 0 for No): ");
                do{
                        isPeakHour = getPeakHour();
                } while (!success);

                // Calculate Base Charge
                if (electricConsumption > 600){
                        rateForEachkWh = 0.546;
                } else if (electricConsumption > 300){
                        rateForEachkWh = 0.516;
                } else if (electricConsumption > 200){
                        rateForEachkWh = 0.334;
                } else{
                        rateForEachkWh = 0.218;
                }
                double baseCharge = electricConsumption * rateForEachkWh;
                totalBill += baseCharge;
                // Calculate surcharge amount
                double surchargeAmount = 0;
                if (isPeakHour){
                        surchargeAmount = baseCharge * 0.10;
                        totalBill += surchargeAmount;
                }

                // Calculate discount
                boolean isDiscount = false;
                if (electricConsumption < 100){
                        totalBill = totalBill * 0.95;
                        isDiscount = true;
                }

                // Calculate rebate amount
                double rebateAmount = 0;
                if (isB40 && totalBill > 40){
                        rebateAmount = 40;
                        totalBill -= 40;
                }

                // Print outputs
                System.out.println("");
                System.out.println("Electric consumption (kWh) : " + electricConsumption);
                System.out.println("User category : " + ((isB40) ? "B40" : "Standard"));
                System.out.println("Peak hours : " + ((isPeakHour) ? "Yes" : "No"));
                System.out.println("Base charge (RM) : " + String.format("%.2f",baseCharge));
                System.out.println("Surcharge charge (RM) : " + String.format("%.2f",surchargeAmount));
                if (isDiscount){
                        System.out.println("Your total consumption is less than 100kWh. You are a Green Saver. : Discount 5%");
                }
                System.out.println("Rebate charge (RM) : " + String.format("%.2f",rebateAmount));
                System.out.println("Total bill (RM) : " + String.format("%.2f",totalBill));
        }
```

```java
        public static int getElectricConsumption(){
                int input;
                try {
                        input = scanner.nextInt();
                } catch (InputMismatchException e){
                        System.out.println("Please enter a integer.");
                        scanner.nextLine();
                        success = false;
                        return -1;
                }
                if (input < 0) {
                        System.out.println("Please enter a positive number.");
                        success = false;
                        return -1;
                }
                success = true;
                return input;
        }

        public static boolean getB40(){
                int input;
                try {
                        input = scanner.nextInt();
                } catch (InputMismatchException e){
                        System.out.println("Please input 1 for B40, 2 for Standard.");
                        scanner.nextLine();
                        success = false;
                        return false;
                }

                if (input == 1) {
                        success = true;
                        return true;
                } else if (input == 2) {
                        success = true;
                        return false;
                } else {
                        System.out.println("Please input 1 for B40, 2 for Standard.");
                        success = false;
                        return false;
                }
        }

         public static boolean getPeakHour(){
                 int input;
                 try {
                         input = scanner.nextInt();
                 } catch (InputMismatchException e){
                         System.out.println("Please input 1 for Yes, 0 for No.");
                         scanner.nextLine();
                         success = false;
                         return false;
                 }

                 if (input == 1) {
                         success = true;
                         return true;
                 } else if (input == 0) {
                         success = true;
                         return false;
                 } else {
                         System.out.println("Please input 1 for Yes, 0 for No.");
                         success = false;
                         return false;
                 }
         }
}
```